Position-based inference with coupled deformable contact

A completed controlled study and the numerical audit that motivates its repair

The model estimates stiffness and friction from observed positions during one squeeze, shear and release, then predicts sliding, edge dropping, curved-obstacle pushing and collision with a moving sphere. This adds force exchange between deformable and rigid bodies while preserving the earlier public-dataset results.

Completed reserved comparison

Method	Complete	World RMSE (mm)	Shape RMSE (mm)
Inferred nonlinear FEM	32/32	13.59	1.78
Rigid + same contact	32/32	20.08	1.91
True-parameter coarse FEM	32/32	12.05	1.61
Adapted PhysTwin spring core	20/32	33.37	1.91
Position ridge	32/32	49.63	7.60

Fixed nominal FEM: 15.73 mm, complete on all 32 cases. Inferred FEM wins 28/32 against rigid contact and 25/32 against nominal FEM; mean world error falls by 32.3% and 13.6%, respectively. On the 20 completed spring-core pairs, inferred FEM averages 15.61 mm and wins 19/20. Repeated interactions share eight objects; 32 trajectories are not 32 independent material samples.

Reference limitation

The original vertex-contact reference failed spatial consistency. A development mesh change from 4 to 6 cells alters edge-drop trajectories by 12.13 and 29.72 mm. The table is therefore evidence about prediction of this frozen discretized simulator, not validated physical ground truth. All cases remain included; 26/32 reserved references pass the separate timestep and penetration checks.

Shape error removes each frame's best rigid transformation. The shape improvement is smaller than the world-position improvement. Missing spring-core trajectories are explicit failures under this study's positive-element-volume guard; its reported mean is conditional on completion.

Algorithm and experimental inputs

1. Represent bodies and contact

The soft block is a conforming tetrahedral mesh with lumped mass. A compressible Neo-Hookean law predicts deformation. The scene contains prescribed tools, static collision geometry and, in one scenario, a dynamic rigid sphere. Penalty normal contact and dissipative, Coulomb-capped viscous friction apply equal/opposite forces and torques. There is no self-contact or continuous collision detection.

For each tetrahedron: $F = D_s D_m^{-1}$, $J = \det(F)$. The strain energy density is $W = (\mu/2)(\mathrm{tr}(F^T F) - 3) - \mu \log(J) + (\lambda/2)\log(J)^2$. Nodal elastic forces are the negative energy gradient. Objective strain-rate viscosity and gravity are added; symplectic Euler advances velocities and positions. Any integration step with $J \le 0.02$ is rejected.

2. Infer from positions, without future object targets

Observe 18 deterministic surface nodes at 100 Hz through the full 0.6-second calibration motion, with independent 0.1 mm Gaussian position noise. Search 7 Young-modulus values by 5 friction values and select the smallest observed-position mean squared error. Initial geometry, density, Poisson ratio, viscosity, gravity, scene geometry and tool trajectories are known. Stiffness and friction are unknown. This is a discrete point estimate, not a calibrated posterior.

3. Transfer parameters to different interactions

Each forecast starts from its supplied initial state. Only fitted physical parameters transfer; the forecast does not continue the calibration state. The original predictor has 2 cells per axis, the reference 4. Predictor timestep is 0.1 ms; reference 0.05 ms with a separate 0.025 ms check for every episode. All four interaction families were used in numerical development. The eight reserved material/friction combinations are new combinations within the search range, not unseen material-range extrapolation.

4. Compare matched explanations

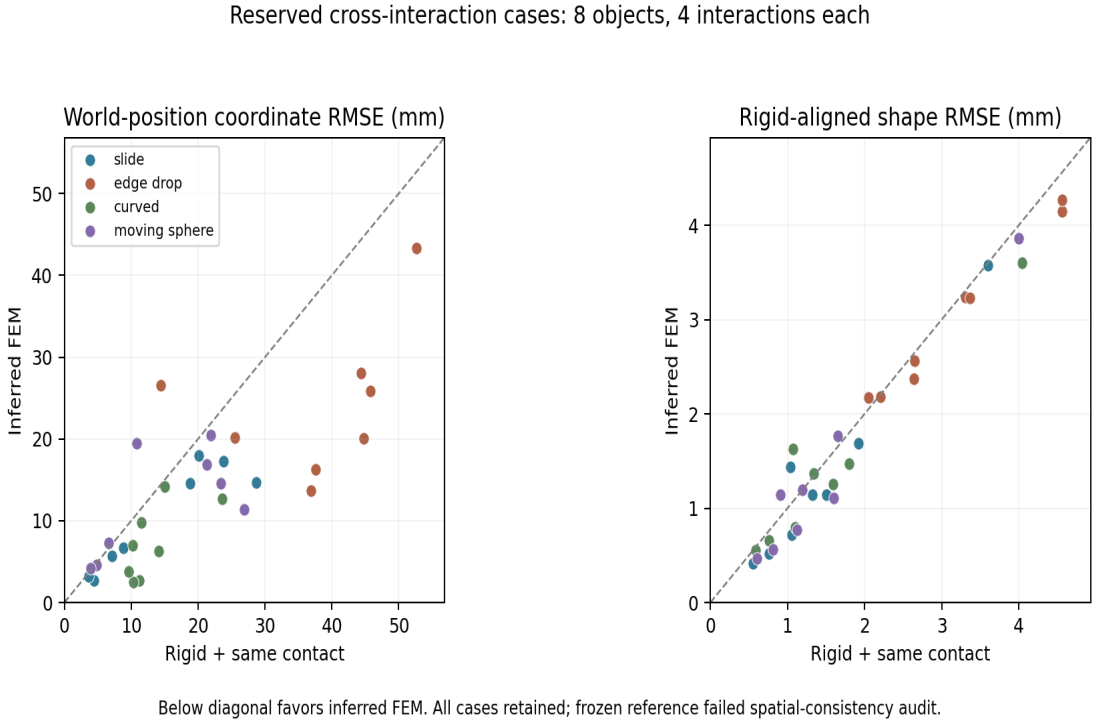
Rigid contact uses the same scene and fits friction. The oracle diagnostic receives true material/friction labels. Fixed nominal FEM uses 6000 Pa and friction 0.35. The native spring baseline uses the authors' unchanged PhysTwin spring-force kernel on this study's graph, masses, contact and integration, with its own effective-scale search. It is not the full authors' method or an official PhysTwin ranking. Position regression lacks collision geometry and is a weaker control.

How much deformation is present?

Scenario	Median nonrigid RMS	Nonrigid / world RMS
Calibration squeeze	1.87 mm	54.8%
Slide	1.03 mm	2.0%
Edge drop	2.67 mm	6.0%
Curved obstacle	1.03 mm	2.2%
Moving sphere	0.99 mm	2.0%

Independent population audit: all 70 references, all 125 fine-grid nodes, uniform coordinate weighting, initial frame excluded, proper rigid fit per frame. Forecast movement is predominantly rigid even though millimetre-scale deformation remains measurable.

Audit, preservation and repair



All 32 reserved cases are shown. Below the diagonal favors inferred FEM. These comparisons retain the failed spatial-reference qualification.

Verified accounting

Independent verification recomputed 3,502 main checks and 1,564 nominal-baseline checks with zero discrepancies. It replayed observation masks, noise, all grid losses, selected hypotheses, saved trajectory scores and qualification flags. The 347 recorded v11/v12 files retain their hashes. Previous SoftVTBench and PhysTwin dataset results are unchanged; no new public-dataset gain is established by this simulation study.

Contact repair is a separate experiment

The new development path integrates the material surface instead of relying on mesh vertices to sample a narrow support. Fixed midpoint sampling still fails the edge-drop world-trajectory gate even at 0.375 mm spacing. Exact integration of clipped plane/box contact footprints is being developed separately, with independent force/torque and friction-integration checks. No fresh test result is claimed here. Reference and predictor resolutions will be selected on development consistency, with the fixed acceptance bounds retained.

Reproducibility and videos

Protocol and results: reports/coupled-contact-v13. Source and commands: scripts/coupled_contact_v13/README.txt. Audits: independent-audit.json, final-protocol-audit.txt, mesh-development-diagnostic.json and motion-population-audit.json. Raw trajectories and datasets remain local. Three video cases were declared before outcome inspection: test0-curved.mp4, test3-moving_sphere.mp4 and test7-edge_drop.mp4. They use common world coordinates, a moving camera, no deformation amplification and explicitly labeled 4x slower playback. The edge-drop clip retains a failed prediction.

Interpretation

The useful progress is explicit two-way contact and improved prediction of a controlled simulator. Accurate physical identification, spatially trustworthy references, calibrated uncertainty, active probing superiority and stronger public-benchmark performance remain separate, unresolved questions.